AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2016

Biology Paper II

Time: 2 hours 25 minutes Marks: 40

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INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are NINE questions. Answer ALL questions. Questions 8 and 9 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

Q.1. (Total 5 Marks)

a. Complete the given table by identifying the branch of biology with respect to its significance. (3 Marks)

Significance of the Branch of Biology	Name of the Branch of Biology
This branch of biology helps scientists to propose appropriate habitat for the living organisms and find out ways to reduce different types of pollution.	
This branch of biology enables scientists to make use of living organisms for the benefits of mankind, e.g. production of bio-fuel.	
This branch of biology supports scientists to find out the evolutionary relationships between different organisms on the basis of dead remains.	

b. Which TWO structural features of a fish enable it to survive in water? (2 Marks)

_____

_____

_____

_____

Q.2. (Total 4 Marks)

a. What would a scientist do if his hypothesis is: (2 Marks)

Supported by experimentation _____

_____

Not supported by experimentation _____

_____

b. 'Use of diagrams in teaching biology improves conceptual understanding of students.'

Draw deduction from the given hypothesis. (2 Marks)

_____

_____

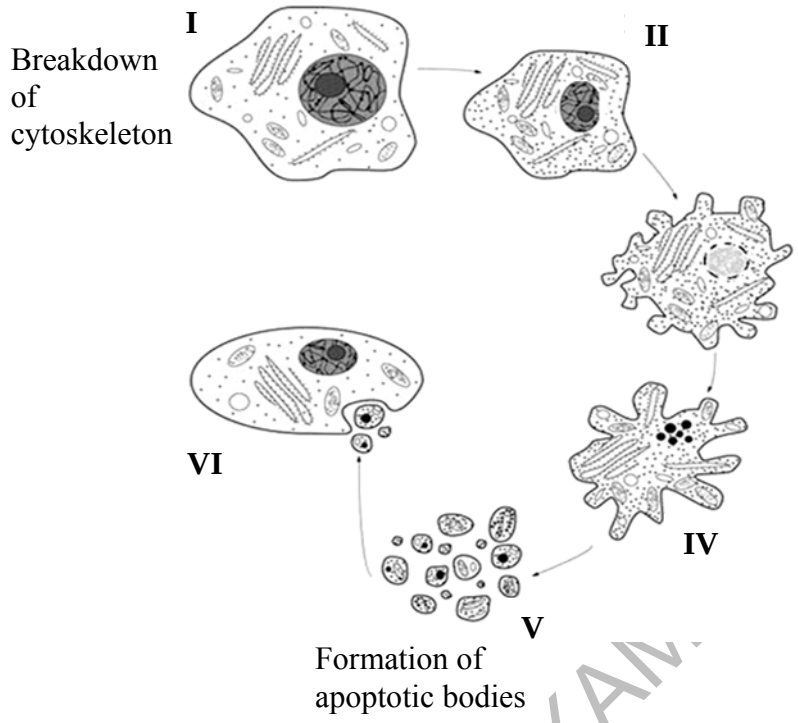
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Q.3. (Total 3 Marks)

The given diagram shows some events of apoptosis. Briefly describe what happens at stages **II**, **IV** and **VI**.



Stage **II**: _____

_____

Stage **IV**: _____

_____

Stage **VI**: _____

_____

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Q.4. (Total 2 Marks)

Hydrogen peroxide is broken down into water and oxygen. The breakdown of hydrogen peroxide can take less time if catalase is used.

Identify the following for the given chemical reaction.

Enzyme: _____

Substrate: _____

Q.5. (Total 3 Marks)

The given reaction takes place in yeast.

$C_6H_{12}O_6 \rightarrow$ **X** + **Y** + ATP

i. Identify products **X** and **Y**. (2 Marks)

_____

_____

ii. In which part of the yeast cell does this reaction occur? (1 Mark)

_____

Q.6. (Total 3 Marks)

Complete the given table.

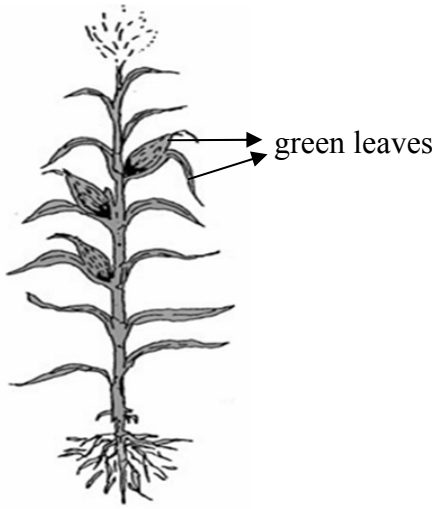
Blood Vessel	Origin	Target Organ
Hepatic Vein		
Renal Artery		Kidneys
Pulmonary Artery	Heart	Lungs

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Q.7. (Total 5 Marks)

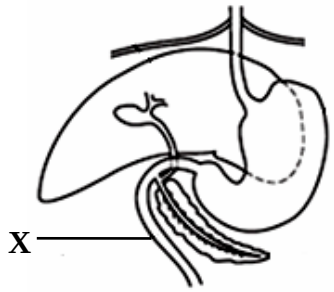
a. The diagram shows a plant which receives appropriate amount of macro-nutrients and micro-nutrients.

Redraw the diagram of the plant showing the effects of deficiency of nitrogen and magnesium. (2 Marks)



Space for drawing

b. The diagram shows some parts of the human digestive system.



Briefly describe how the THREE enzymes entering into region **X** of the alimentary canal help in the digestion of food. (3 Marks)

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Q.8. (Total 7 Marks)

EITHER

a.

i. Define anatomy.

ii. Explain how studying immunology and morphology help scientists in the prevention and treatment of various diseases.

OR

b.

i. What are meristematic tissues?

ii. Describe structural adaptations in any THREE of the permanent tissues in plants that help to perform their function efficiently.

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Q.9. (Total 8 Marks)

EITHER

a. Describe the role of chlorophyll, water and NADPH in the conversion of light energy to chemical energy in plants.

OR

b. The table below gives information about the donor's and recipient's blood groups in case of two different blood transfusions.

Blood Transfusion	Donor's Blood Group	Recipient's Blood Group
Case I	A	AB
Case II	B	O
Case III	O^+	?

i. Explain the results of blood transfusion in cases **I** and **II** shown above.

ii. What are the possible recipients of O^+ blood in case **III**? Give a reason to support your answer.

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